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Unlocking the Potential of Pseudo-Labels: A Visual Anchoring Framework for Cross-Domain Behavior Annotation

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ABSTRACT As behavioral analysis across application domains scales up, the demand for annotated video data grows exponentially, presenting a significant bottleneck. This study introduces an automated visual anchoring framework utilizing general-purpose Vision-Language Models (VLMs) and Large Language Models (LLMs) to reliably annotate out-of-distribution video datasets. Specifically, we investigate whether generic human behavior representations can correctly guide annotation on specialized domains. By breaking down annotation tasks into single-frame conceptual alignment and multi-frame temporal reasoning, we evaluate zero-shot annotation generation without domain-specific re-training. Our findings confirm that when visual behaviors are sufficiently grounded in distinct frame-level states, pseudo-labels reach near-ground-truth quality. Pseudo-label fine-tuned models achieved 88.65% accuracy on BowTurnHead and 43.22% on TeacherBehavior. Selective annotation routing further mitigates distribution shifts, improving robust behavior recognition.

INDEX TERMS Action Recognition, Large Language Models, Pseudo-Labeling, Video Understanding

I. INTRODUCTION

THIS paper demonstrates the utility of VLMs in cross-domain annotation.